

Engineering Department Doc, Layer 2 v0.2.1 team_width

Warning Review Result

Soccer TIPS: Tactical Intelligence & Performance System

Purpose: This addendum closes the Layer 2 v0.2.1 team_width warning review. It documents the decision from the local generated outputs and preserves the scope boundary: no Layer 3 modeling, tactical identity profiling, AI interpretation, dashboards, reports, recommendations, or advanced tactical claims were created.

1. Status Decision

Item	Decision
Status decision	requires code/definition patch
Cause assessment	width/depth semantic mismatch / code definition issue
Layer 3 recommendation	patch before use; exclude team_width from Layer 3 v0.1 tactical claims until patched and revalidated
Implementation recommendation	patch metric definition, rerun Layer 2 batch, rerun Layer 2 QA, rerun football sanity validation, and rerun this review

General Developer 1 note: The first local review script conservatively labelled the metric unsafe/unresolved. After inspecting the raw-coordinate evidence and exact plots, the stronger finding is that the current output appears to have a width/depth semantic mismatch. The output value labelled team_width behaves like the longitudinal x-range, while the lateral y-range appears under team_depth for the worst examples.

2. Warning Summary

Field	Value
Total frame-team rows reviewed	873,296
Violating rows	95,324
Violating row percentage	10.915428%
Max team_width	92.690
Median team_width	52.660
Pitch-width bound used	68.000
Affected matches	1886347, 1899585, 1925299, 1953632, 1996435, 2006229, 2011166, 2013725, 2015213, 2017461
Affected teams	1802, 1803, 1804, 1805, 2380, 4177, 866, 867, 868, 869, 870, 871
Affected periods	1, 2
Affected in-possession phases	build_up, chaotic, create, direct, finish, quick_break, set_play, transition
Affected out-of-possession phases	chaotic, defending_direct, defending_quick_break, defending_set_play, defending_transition, high_block, low_block, medium_block
Affected possession statuses	in_possession, out_of_possession, unknown

3. Worst Examples Table

match	frame	period	team	phase	possess	team_	team_d	players	detecte	extrap
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					ion	width	epth		d	olated
2017461	55698	2	4177		unkno wn	92.69	34.24	11	2	9
2017461	55695	2	4177		unkno wn	92.69	34.25	11	1	10
2017461	55696	2	4177		unkno wn	92.69	34.26	11	1	10
2017461	55694	2	4177		unkno wn	92.68	34.2300 000000 000004	11	1	10
2017461	55697	2	4177		unkno wn	92.68	34.27	11	2	9

The full worst-examples table, including phase labels, hull area, average distance, and notes, is saved separately as layer2_v021_team_width_worst_examples.csv.

4. Raw Coordinate Inspection

match_id	frame	team_id	min y	max y	actual y range	outside y count	min-y player	min detect ed	max-y player	max detect ed	one-player flag
2017461	55698	4177	-22.57	11.67	34.24	0	133501	False	133498	False	True
2017461	55695	4177	-22.67	11.58	34.25	0	133501	False	133498	False	True
2017461	55696	4177	-22.63	11.63	34.26	0	133501	False	133498	False	True
2017461	55694	4177	-22.71	11.52	34.23	0	133501	False	133498	False	True
2017461	55697	4177	-22.61	11.66	34.27	0	133501	False	133498	False	True

Key finding: in the worst examples, player y-coordinates remain inside the expected pitch y bounds and the actual y range is about 34.23 to 34.27 meters. That value matches the output column team_depth, not the output column team_width. This strongly indicates that the current team_width/team_depth labels or calculations are reversed relative to the documented football convention.

5. Time-Continuity Inspection

match_id	frame	team_id	current team_width	neighbor rows	violating frames in window	one-frame spike	persists	window median	related metrics spike
2017461	55698	4177	92.69	21	21	False	True	92.50	False
2017461	55695	4177	92.69	21	21	False	True	92.50	False
2017461	55696	4177	92.69	21	21	False	True	92.50	False
2017461	55694	4177	92.68	21	21	False	True	92.50	False
2017461	55697	4177	92.68	21	21	False	True	92.50	False

The warning is not a one-frame spike. The worst examples persist across neighboring frames. However, the persistence appears to be persistence of a likely mislabeled longitudinal spread, not proof that lateral team width exceeds the pitch width.

6. Visual Evidence

Item	Result
Existing layer2_sanity plots found	11
Exact outlier plots created	5
Review plot folder	C:\Users\amsch\. Personal Work\Projects\Soccer TIPS\outputs\figures\layer2_sanity\team_width_review

Visual review of the provided exact flagged-frame plots supports the raw-coordinate finding: the plotted lateral y-range is within pitch bounds, while the large spread is primarily along the pitch length/x-axis. Team separation, player positions, centroids, and hulls look structurally plausible, but the label team_width is not safe because it appears to be measuring depth/longitudinal spread in the current output.

7. Phase-Segment Impact

Item	Result
Flagged phase segments found	1559
Summary columns checked	team_width_mean, team_width_median, team_width_std, team_width_min, team_width_max
Interpretation	Flagged rows appear in many phase summaries, so the issue can materially affect phase-level team_width aggregates if not patched before Layer 3.

8. Final Recommendation

- Do not use team_width for Layer 3 v0.1 tactical claims yet.
- Patch the metric definition or column assignment so team_width = max(y_metric) - min(y_metric) and team_depth = max(x_metric) - min(x_metric), matching the documented convention.
- After patching, rerun Layer 2 batch, Layer 2 QA, football sanity validation, and this team_width review.
- Until the patch is accepted, keep the existing team_width output as diagnostic only and treat phase-level team_width summaries as not trusted for tactical interpretation.

9. Scope Guard

- No Layer 3 outputs were created.
- No tactical identity profiles were created.
- No AI interpretation, dashboards, reports, or recommendations were created.
- No final tactical claims should be made from team_width until the Lead Developer approves the metric patch and validation rerun.